

Development and implementation of a work productivity and activity impairment assessment process for employed patients with migraine in a health-system specialty pharmacy setting



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Background

- Employees diagnosed with migraine miss an average of 4.4 workdays each year and have another 11.4 reduced productivity workdays due to migraine attacks.¹ This productivity lost to migraines costs US employers billions of dollars annually.^{2,3}
- Health-system specialty pharmacy (HSSP) teams support patients in managing migraines through ongoing monitoring of treatment effectiveness and safety, while also ensuring access to medications and promoting long-term adherence.
- Although HSSP teams are well-positioned to support the clinical management of migraine, the integration of standardized work productivity and impairment assessment tools into routine patient care workflows remains limited.

Objective

To describe a work productivity and activity impairment assessment process for employed patients with migraine in an HSSP setting.

Methods

HSSP Quality Improvement Initiative

- The Work Productivity and Activity Impairment: Specific Health Problem V2.0 (WPAI:SHP) questionnaire was integrated into the pharmacy workflow in February 2024 through an existing clinical management protocol at a single HSSP.
- The WPAI:SHP is composed of six questions and yields scores for absenteeism (percent of work time missed), presenteeism (percent impairment at work), overall work productivity loss (percent of overall work impairment/absenteeism plus presenteeism), and activity impairment (percent of activity impairment).
- Full-time employed adults with a newly prescribed targeted migraine specialty therapy were eligible to complete the WPAI:SHP. For participants, the questionnaire was collected once per month for six months.

Data Analysis

- Wilcoxon signed-rank tests were used to compare scores at baseline and three months, as well as at baseline and six months.
- A p-value <0.05 was considered statistically significant.
- Pharmacists who administered the WPAI:SHP were given a survey at the conclusion of the study to assess its utility in the management of patients with migraines.

Results

In total, 18 patients were enrolled and 2 were lost to follow-up. Most patients were female (77.8%), and the median age was 39.5 years [IQR, 15].

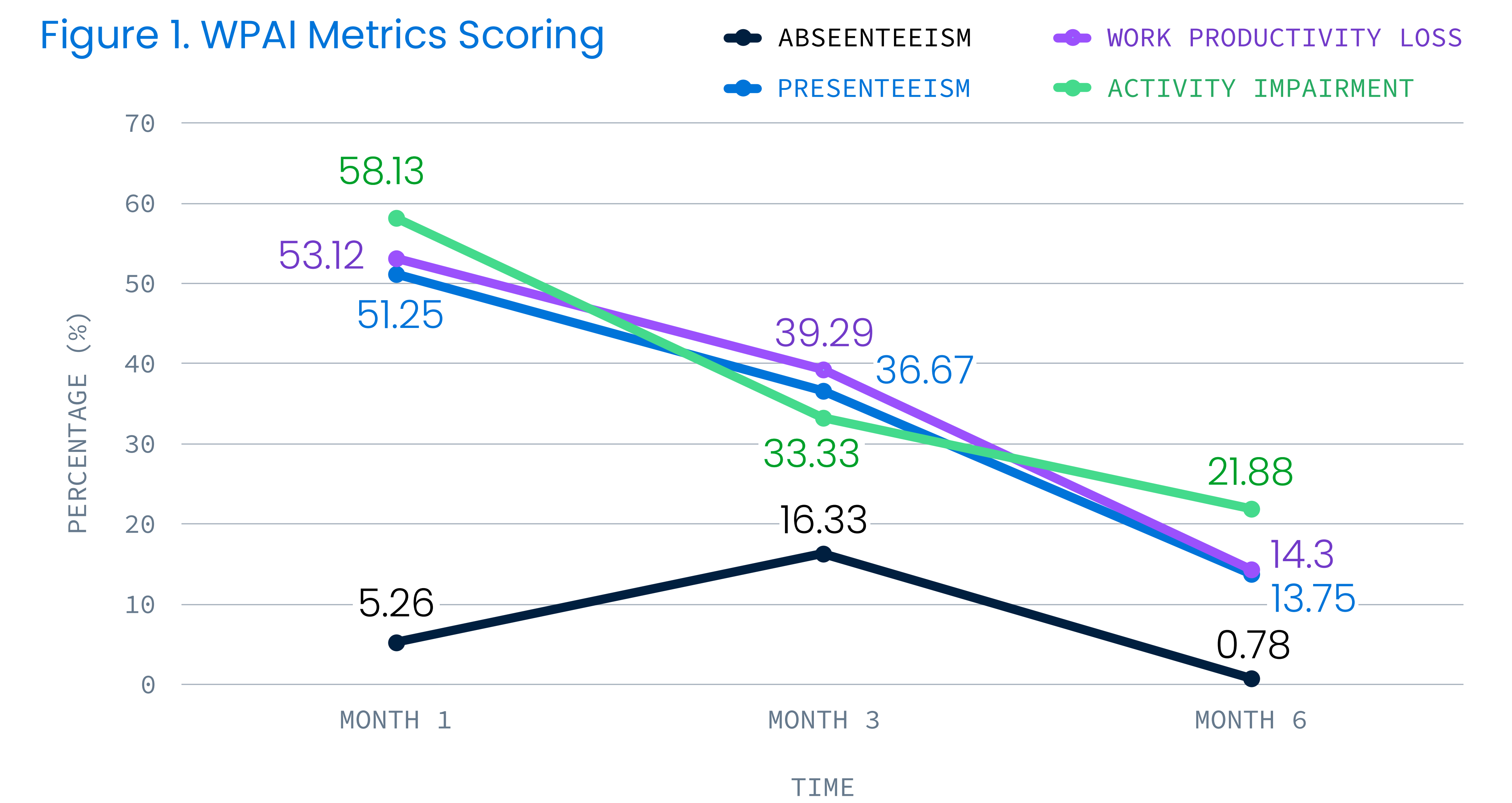


Table 1. Percentage Change of WPAI Metrics

Metric	Percentage Change	
	Month 1 to Month 3	Month 1 to Month 6
Absenteeism	210.5% increase (p=1.00)	85.2% decrease (p=0.10)
Presenteeism	24.4% decrease (p=0.08)	73.2% decrease (p=0.001)
Work Productivity Loss	26.0% decrease (p=0.06)	73.1% decrease (p=0.002)
Activity Impairment	42.7% decrease (p=0.04)	63.4% decrease (p=0.02)

Results

Table 2. Pharmacist Feasibility Survey (n=3)

Question (True/False)	Response (%)
The WPAI:SHP was easy to use (T/F)	100% True
The WPAI:SHP provided information that was useful during the assessment of patients with migraine (T/F)	100% False
I would like to incorporate the WPAI:SHP in future patient assessments (T/F)	100% False

Conclusion

- The WPAI:SHP can help assess how HSSP clinical management for migraines affects patients' work productivity and daily activities. However, in this HSSP setting, pharmacists did not believe the questionnaire provided additional value beyond the clinical data already being collected as part of the existing clinical management protocol.

Limitations

- A small sample size of only 18 patients limits statistical power and generalizability.
- The single-center setting and only three pharmacists providing feedback on the feasibility of WPAI into workflow both restrict the ability to draw meaningful conclusions across diverse pharmacy settings.
- Pharmacist feasibility assessment was limited to three items and did not provide detail about time burden, workflow disruption, or impact on patient care.

References

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